

Double Integrals



Double integrals over a rectangle

- 1 Set up the double integral of $f(x, y) = xy$ over the rectangle $[0, 2] \times [0, 3]$ (integrate y first).
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Iterated integrals

- 2 Evaluate $\int_0^2 \int_0^3 xy \, dy \, dx$.
 - 3 Evaluate $\int_0^1 \int_0^2 (x^2 + y) \, dy \, dx$.
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Fubini's Theorem: order of integration

- 4 Verify Fubini's Theorem for $f(x, y) = xy$ on $[0, 1] \times [0, 2]$ by evaluating the integral in both orders.
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Double integrals over general regions

- 5 Evaluate $\int_0^1 \int_0^x (x + y) \, dy \, dx$.
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Polar coordinates

- 6 Convert $\iint_D (x^2 + y^2) \, dA$ to polar coordinates, where D is the disk $x^2 + y^2 \leq 4$, and evaluate.
- 7 Evaluate $\iint_D xy \, dA$ where D is the quarter disk $x^2 + y^2 \leq 1$, $x \geq 0$, $y \geq 0$, using polar coordinates.